

Juliana C. Taube

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Education

Georgetown University

PHD IN BIOLOGY

Advisor: Shweta Bansal, PhD

Washington, DC

2023 - present

Bowdoin College

A.B. *summa cum laude* WITH HONORS IN MATHEMATICS, MINOR IN BIOLOGY

Undergraduate thesis: Modeling coupled disease-behavior dynamics of SARS-CoV-2 using influence networks

Advisors: Mary Lou Zeeman, PhD (Mathematics) and Mohammad Irfan, PhD (Computer Science, Digital & Computational Studies)

Brunswick, ME

2017 - 2021

Research Experience

Georgetown University

GRADUATE STUDENT RESEARCHER, ADVISOR: SHWETA BANSAL, PhD

- Described spatiotemporal trends in self-reported non-household contacts
- Characterized gender & race/ethnicity composition & citation practices in the field of infectious disease dynamics
- Disentangling aerosol and droplet respiratory disease transmission routes
- Quantifying the social and disease effects of hygienic behaviors across species & scales

Washington, DC

2023 - present

RESEARCH ASSOCIATE

- Analyzed spatiotemporal trends in self-reported mask-wearing behavior during the U.S. COVID-19 pandemic
- Mapped the immune landscape of orthopoxviruses at a fine spatial resolution using demographic and vaccination data

2021 - 2023

Bowdoin College

STUDENT RESEARCHER, ADVISORS: MARY LOU ZEEMAN, PHD; MOHAMMAD IRFAN, PHD

- Continued thesis work on coupled disease-behavior model by varying scales of risk perception & contact network structure

Remote

2021

Centers for Disease Control and Prevention, Division of Global Migration & Quarantine, Office of Innovation, Development, Evaluation, and Analytics

INTERN, ADVISORS: ARDATH GRILLS, PHD; SARAH BOWDEN, PHD; MICHAEL JOHANSSON, PHD

- Organized county census & meat-packing plant data for boosted regression tree model to predict & characterize COVID-19 hotspots
- Collected data on COVID-19 superspreader events at large gatherings to estimate dispersion parameter
- Contributed to model implementation and assessment of interventions to mitigate COVID-19 spread on cruise ships

Remote

Summer 2020

University of Georgia, Odum School of Ecology

STUDENT RESEARCHER (REMOTE), ADVISORS: JOHN M. DRAKE, PHD; PAIGE B. MILLER, PHD

- Expanded transmission trees database to include COVID-19, extended analysis, and released data online

Athens, GA

Fall 2020

STUDENT RESEARCHER (REU)

- Compiled and standardized infectious disease transmission trees from the literature into an R database
- Analyzed predictors of outbreak size & characterized the role of superspreading across outbreaks

Summer 2019

Dartmouth Hitchcock Medical Center

INTERN, ADVISOR: PETER F. WRIGHT, MD

- Compiled and summarized literature on smallpox and polio eradication efforts, highlighting similarities and differences
- Organized data on bronchiolitis and RSV hospitalizations in New England

Lebanon, NH

Summer 2017

Publications

6. **Taube JC**, Susswein Z, Colizza V, Bansal S. (2025). Characterising non-household contact patterns relevant to respiratory transmission in the USA: analysis of a cross-sectional survey. *The Lancet Digital Health*, 7(8), 100888. doi:10.1016/j.landig.2025.100888  

5. **Taube JC***, Merritt A*, Bansal S. (2025). The extent and drivers of gender and race/ethnicity imbalance in infectious disease dynamics research. *Proceedings of the Royal Society B*, 292(2050), 20250960 doi: 10.1098/rspb.2025.0960   
4. **Taube JC**, Susswein Z, Bansal S. (2023). Spatiotemporal trends in self-reported mask-wearing behavior in the United States: Analysis of a large cross-sectional survey. *JMIR Public Health and Surveillance*, 9, e42128. doi: 10.2196/42128   
3. **Taube JC***, Rest EC*, Lloyd-Smith JO, Bansal S. (2022). The global landscape of smallpox vaccination history and implications for current and future orthopoxvirus susceptibility: a modelling study. *The Lancet Infectious Diseases*, 23(4), 454-462. doi: 10.1016/S1473-3099(22)00664-8.   
2. **Taube JC**, Miller PB, Drake JM. (2022). An open-access database of infectious disease transmission trees to explore superspreader epidemiology. *PLoS Biology*, 20(6), e3001685. doi: 10.1371/journal.pbio.3001685.   
1. Wright PF, Hoen AG, Jarvis JD, Zens MS, Dade EF, Karagas MR, **Taube JC**, Brickley EB. (2022). Bronchiolitis hospitalizations in northern New England: Clues to disease prevention. *Therapeutic Advances in Infectious Disease*, 9, 1-11. doi: 10.1177/ 20499361221099447  

* Authors contributed equally

Preprints & In Prep

2. Simin PT, **Taube JC**, Vergu E, Bansal S, Opatowski L, Poletto C. Contact structure and population immunity shape the selective advantage of emerging variants. *medRxiv*. doi: 10.1101/2025.11.07.25339691. 
1. Woika CM*, **Taube JC***, Colizza V, Bansal S. Characterizing the impacts of disease on behavior: policy, perception, and potential for infection. *medRxiv*. doi: 10.64898/2026.03.09.26347630. 

Presentations

Do humans behave rationally? Assessing potential drivers of US pandemic behavior change

Talk for the EPIcx Lab at INSERM

2026

Putting network epidemiology theory to the test: Estimating contact structure across epidemic conditions with implications for control

Invited talk at the Spread of Contagions in Networks NetSci 2026 satellite

2026

From proximity to pandemics: How human behavior and disease spread interact

Graduate Research Seminar for the Georgetown Biology Department

2026

Characterizing contact patterns and the drivers of pandemic behavior change in the US

Talk for the EPIcx Lab at INSERM

2025

All contacts are not equal: Spatiotemporal constraints shape network structure

Lightning Talk and Poster at NetSci 2026

2026

Poster at Epidemics 10

2025

Characterizing the impacts of disease on behavior: policy, perception, and potential for infection

Talk at MIDAS Meeting 2025

2025

The global landscape of smallpox vaccination history and implications for current and future orthopoxvirus susceptibility

Poster at World Vaccine Congress 

2025

Rapid Talk at MIDAS Meeting 2024 

2024

The extent of gender & race/ethnicity imbalance in infectious disease dynamics research

Invited VERENA Lighthouse Lightning Talk (virtual)

2024

Contact patterns in the US vary over space but not time

Poster at MIDAS Meeting 2023 

2023

Poster at EEID 2023

2023

Characterizing spatiotemporal trends in self-reported mask-wearing behavior in the U.S.

Talk at MIDAS Meeting 2022 

2022

Talk at Delphi's COVID-19 Trends and Impacts Survey Monthly Collaboration Meeting (virtual)

2022

Modeling coupled disease-behavior dynamics of SARS-CoV-2 using influence networks

Invited talk for Prof. Mohammad Irfan's Research Group at Bowdoin College (virtual)

2021

An open-access database of infectious disease transmission trees to explore superspreader epidemiology

Rapid Fire Talk at MIDAS Meeting 2021 (virtual) 

2021

Poster at Epidemics 7 

2019

Awards & Honors

Total \$130,055

2026 **NetSci 2026 Best Lighting Talk**

2025 \$350 **Georgetown Conference Travel Grant** to attend Epidemics 10

2024 \$2,785 **SISMID Halloran Scholarship** and travel funding

2023 \$111,000 **NSF Graduate Research Fellow**

2021 \$1,920 **Student Faculty Research Grant Fellowship** for summer research funding

2020 **Phi Beta Kappa**

2020 \$5,000 **Bowdoin Funded Internship Grant** for internship with CDC

2019 \$2,000 **REU Travel Grant** from Rocky Mountain Biological Laboratory to attend Epidemics 7

2019 **Bowdoin College Goldwater Scholarship Nominee**

2017 \$3,000 **Bowdoin Faculty Scholar** for excellence in high school courses

2017 \$4,000 **Bowdoin National Merit Scholarship**

Teaching Experience

Georgetown University

Washington, DC

MENTOR

Fall 2023 - Spring 2026

- Supervised an undergraduate student's research project, including senior thesis

BIOLOGY TEACHING FELLOW

Fall 2022, 2023, 2024

- Modeling Populations and Diseases (BIOL 422/4615, GLID 522/5522): responsible for leading the weekly computational lab; grading labs, homeworks, & papers; holding office hours; & creating/updating keys for all assignments

Bowdoin College

Brunswick, ME

MATHEMATICS TEACHING ASSISTANT

Spring 2021

- Partial Differential Equations (MATH 3209): created videos to explain homework solutions or review confusing concepts

DIGITAL AND COMPUTATIONAL STUDIES TEACHING ASSISTANT

Spring 2021

- Contagion (DCS 3350): held office hours; compiled flight, mobility, population, and contact tracing data for student projects

COMPUTER SCIENCE TEACHING ASSISTANT

2019 - 2020

- Introduction to Computer Science (CSCI 1101, 4 semesters): held weekly office hours for multiple sections
- Social and Economic Networks (CSCI/DCS 2350, Fall 2020): held weekly office hours

Service

Georgetown Biology Organization of Graduate Students Secretary

Washington, DC

Take notes & communicate decisions to graduate students, manage student listserv & shared calendar

2026-2027

MIDAS Trainee Executive Committee

Member in 2023, co-lead in 2024 & 2025; planned & facilitated trainee events for the MIDAS Network, including skill workshops, career panels, research talks, & networking sessions at the annual conference

Remote

2023 - 2025

Georgetown Biology Student-Faculty Liaison

Took notes at department faculty meetings and relayed important information to Biology graduate students

Washington, DC

2024 - 2025

Bowdoin Curriculum Implementation Committee

Alternate member; attended committee meetings to provide student perspective on course proposals

Brunswick, ME

2020 - 2021

Training

Apprenticeship in Teaching Program

GEORGETOWN UNIVERSITY

Attended pedagogy workshops to develop teaching skills, including syllabus & course design, & teaching with AI

Washington, DC

2023 - 2025

Summer Institute in Statistics and Modeling in Infectious Diseases (SISMID)

EMORY UNIVERSITY

Attended MCMC I, MCMC II, and Simulation-based Inference for Epidemiological Dynamics modules

Atlanta, GA

Summer 2024

Studying Identity, Diversity, and Equity in Science

GEORGETOWN UNIVERSITY

Discuss identity in science from a variety of disciplines to identify how research can contribute to a more equitable society

Washington, DC

Summer 2023

Teaching, Learning & Innovation Summer Institute

GEORGETOWN UNIVERSITY

Workshops on universal design, assessments in STEM courses, use of office hours, and science communication

Washington, DC

Spring 2022

Exploring Racial Justice: An Intragroup Dialogue for White Identified Students

BOWDOIN COLLEGE

Six-week workshop exploring social justice issues and white privilege without burdening students of marginalized identities

Virtual

Spring 2021

Summer Institute in Biostatistics

UNIVERSITY OF MINNESOTA

Coursework in biostatistics, epidemiology, and statistical computing using R and SAS

Minneapolis, MN

Summer 2018

Skills

Programming R, Python, Stan, SQL

Software LaTeX, Git, MacOS, Microsoft Office